



OPEN A cyanobacteria-mediated TiO₂ nanoparticle and propolis cream enhances antifungal activity and wound healing against *Candida albicans* in mice

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Nanoformulations of natural drugs derived from cyanobacteria can be designed using various nanoparticles, thereby acquiring suitable properties for antimicrobial, cytotoxic, and antioxidant activities. In this study, we aimed to use a combination of natural nanoparticles and propolis to produce a wound healing cream and evaluate its antifungal, cytotoxic, and antioxidant activities through various tests. Four types of creams were prepared: a base cream, a cream containing nanoparticles, a cream containing propolis, and a cream containing both propolis and nanoparticles. The nanoparticles were polyvinyl alcohol/titanium dioxide nanoparticles (PVA/TiO₂ NCs) extracted from the cyanobacterium *Neowestiellopsis persica*. The results of measuring the diameter of inhibition zones, minimum inhibitory concentration (MIC), and minimum fungicidal concentration (MFC) of the prepared creams showed that *Candida albicans* is much more resistant than *Trichophyton rubrum*. Histological examination of wound samples on day 14 revealed the highest epithelialization and the lowest levels of neutrophils and macrophages in mice treated with the cream containing both propolis and nanoparticles (group 4). Overall, the cream prepared with nanoparticles and propolis demonstrated significant improvement in wound healing compared to the control group. This study is the first to show the effectiveness of a therapeutic cream based on natural ingredients in promoting wound healing in animals.

Keywords *Neowestiellopsis persica*, Cyanobacteria, Titanium dioxide nanoparticles, Propolis, *Candida albicans*, Artificial infection

Abbreviations

PVA/TiO ₂ NCs	Polyvinyl alcohol/titanium dioxide nanoparticles
MIC	Minimum inhibitory concentration
MFC	Minimum fungicidal concentration
H&E	Hematoxylin and eosin
DMSO	Dimethyl sulfoxide
CCC	Cyanobacteria Culture Collection
SDA	Sabouraud dextrose agar
TEM	Transmission Electron Microscopy
DPPH	2,2-diphenyl-1-picrylhydrazyl
NCBI	National Cell Bank of Iran
MTT	3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide

Wounds are defined as a disruption in tissue integrity. Wounds are classified in various ways and may be categorized as superficial, partial, or full-thickness based on the number of skin layers and the affected area. Wounds can be clean or infected, acute or chronic¹.

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Acute wounds typically heal completely within an expected timeframe, usually 8 to 12 weeks, with minimal scarring, while chronic wounds heal slowly over longer periods exceeding 12 weeks and often recur. Such wounds fail to heal due to repeated tissue damage or other physiological conditions such as diabetes and malignancies, persistent infections, inadequate initial treatment, and other patient-related factors. Chronic wounds include pressure ulcers and venous leg ulcers. Some chronic wounds also originate from acute wounds. They are often referred to as non-healing wounds and require specialized care².

Since 2.45 billion years ago, algae have developed extensive diversity for their survival in competitive ecological niches. This adaptation has led to the production of a wide range of diverse biochemical compounds. Polysaccharides, cell wall structures, mucopolysaccharides, exopolysaccharides, and storage polysaccharides in algae are recognized as important biomaterials for medical and pharmaceutical applications. These polymers, composed of simple sugars (monosaccharides) linked by glycosidic bonds, provide highly diverse structures and thus serve as an exceptional source for drug discovery³.

The most commonly used wound-healing polysaccharides are extracted from marine algae, particularly macroalgae, which contain 4 to 76% polysaccharides per unit dry weight. Cell wall polysaccharides mainly consist of cellulose, hemicellulose, and neutral polysaccharides. They have numerous commercial applications in products such as stabilizers, thickeners, emulsifiers, food, feed, beverages, and more. Green algae contain sulfated polysaccharides, sulfated galactans, and xylans, while brown algae possess alginic acid, fucoidan (sulfated fucose), laminarin (β -1,3-glucan), and sargassan. Red algae produce agars, carrageenans, xylans, floridean starch (amylopectin-like glucan), water-soluble sulfated galactans, and also porphyran as mucopolysaccharides located in intercellular spaces. Hydrophilic polysaccharides have the ability to absorb large amounts of water, salts, or physiological solutions, which promotes moist wound healing. Additionally, due to their bioactivity, these products may play a significant role in the wound healing process⁴.

Nanoformulations of natural drugs derived from cyanobacteria can be designed using various nanoparticles to achieve suitable properties for antimicrobial, cytotoxic, and antioxidant activities⁵. For example, nanoparticles can be constructed from materials of diverse origins, inorganic (such as metals, carbon, silica, and their respective oxides) and organic (such as polymers and lipids). Moreover, nanoparticle size can be manipulated to range from a few nanometers to over one millimeter, their shape can be smooth or edged, their form can be rigid or soft, and their surface can be functionalized with various molecules. The greatest advantage of carrier nanoparticles for target anticancer agents is that these particles can enhance drug efficacy through several remarkable mechanisms, including increased solubility for better drug delivery, extended drug half-life in circulation, increased drug accumulation in target cancerous tissues or cells, sustained and steady drug release, and reduced drug resistance⁶. All these factors have led to increasing access to bioactive anticancer compounds based on microalgae and commercialized microalgal drugs. In the near future, biocompatible methods using cyanobacterial extracts will be developed as potential sources of reducing and stabilizing agents for green nanoparticles. It is hoped that in the near future, more clean, biocompatible, safe, and cost-effective synthesis methods will also be utilized. Additionally, nanoparticle stabilization in various matrices should follow precise and judicious methods that lead to sustained release of the antibacterial agent. The optimal solution should ensure localized activity of the antibacterial agent with minimal release into the environment. Furthermore, other researchers have demonstrated in a study that biogenic silver nanoparticles are more effective antibacterial agents compared to chemical silver nanoparticles. Secondly, antibacterial nanoparticles may bring fundamental changes to the pharmaceutical industry. Using nanoparticles, the solubility of poorly water-soluble antibiotics will be improved⁷.

Thirdly, the cellular toxicity of antibacterial nanoparticles toward bacteria as well as humans must be considered, as most heavy metal ions (Ag^+ and Cu^{+2}) are toxic. Silver ions are toxic to humans at very high doses. In most cases, silver ions exhibit effective antibacterial activity at low concentrations of approximately 0.001 to 0.05 ppm⁸. It is expected that in the near future, antibacterial nanoparticles based on environmentally safe materials such as alkali metals (magnesium and calcium) will be developed. Additionally, TiO_2 is a material that, despite being safe, has high potential (used globally as an excipient in pharmaceuticals and as a colorant in the food industry) and is light-sensitive. Future research should focus on reducing the activation energy of TiO_2 nanoparticles to exhibit catalytic activity in the visible light range rather than UV. However, it should be noted that while bulk TiO_2 , which has widespread applications, is highly biocompatible, nano-scale TiO_2 has been reported as toxic in some studies⁹. Furthermore, toxicity of TiO_2 nanoparticles has also been demonstrated in *in vivo* studies relative to mouse models. Accordingly, the potential impact of TiO_2 nanoparticles on living organisms should be investigated due to their small particle size and high specific surface area. Moreover, other researchers have shown in a study that TiO_2 nanoparticles without photoactivation had no effect up to 100 $\mu\text{g/mL}$, and UV light irradiation led to a significant increase in toxicity induced by the nanoparticles¹⁰. As observed, although silver nanoparticles are effective antibacterial agents, their cellular toxicity must always be considered, whereas nanoparticles based on alkali metals are non-toxic and a suitable replacement for them. It is predicted that new antibacterial nanoparticles will be based on biocompatible and safe materials possessing physicochemical properties and nanoscale features¹¹. The widespread use of nanoparticles in everyday consumer products may pose a threat to the environment. Other researchers have reported the detection of silver sulfide nanoparticles in sewage sludge at the final stage of urban wastewater treatment plants. They stated that the presence of silver nanoparticles in consumer products leads to increased silver ions in wastewater, which, in the presence of reduced sulfur ions and anaerobic conditions, results in the formation of silver sulfide nanoparticles. Furthermore, the release of silver nanoparticles into the environment is likely a threat to a wide range of living organisms¹².

Propolis, or bee glue, is the world's strongest natural antibiotic without side effects and possesses numerous therapeutic properties. Among the most important therapeutic properties of propolis are its antimicrobial and antitumor effects, immune system enhancement, suitability for skin and teeth, benefits for the gut and uterus,

improved lung function, wound healing and cataract treatment, assistance in treating urinary tract infections, and its biological properties. Propolis contains polyphenols such as flavonoids, which are a type of antioxidant considered among its most important compounds. Flavonoids are plant compounds that exhibit high levels of antioxidant activity¹³.

Phenolics are compounds produced by plants that protect them against oxidative stress. Both flavonoids and phenolics are found in most fruits and vegetables and have been attributed to many health benefits in plant-based diets. Another important component of propolis is the enzymes it contains, which are a combination of salivary secretions and enzymatic materials. These compounds accelerate various enzymatic reactions, enhance cell metabolism, improve blood circulation, and promote collagen fiber formation¹⁴.

Although numerous studies have focused on the development of microalgal nanoformulations and their awareness in medical sectors to date, no formulations have been developed for preparing wound-healing compounds using natural nanoparticles produced by cyanobacteria in combination with propolis. In fact, only some secondary metabolites of microalgae and cyanobacteria and photoprotective compounds have been used in the form of anti-aging creams, antioxidants, and anti-inflammatory creams and drugs isolated from these photosynthetic microorganisms. Since examining compounds effective against chronic and acute wounds isolated from microalgal species for use in nanodrugs toward efficient commercialization requires greater attention and importance, this study aimed to use a combination of natural nanoparticles and propolis to produce a wound-healing cream and evaluate its antimicrobial, cytotoxic, and antioxidant activities through various tests.

Materials and methods

Cultivation of *Neowestiellopsis persica* strain

The *N. persica* cyanobacterium strain was isolated and cultured from the Cyanobacteria Culture Collection (CCC) at the Science and Research Branch of Islamic Azad University. Purification and axenization of cultures were performed in 400 mL of liquid Z8 medium in a growth chamber at 28 °C with continuous fluorescent lighting at an intensity of 300 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ for 10 to 15 days¹⁵ (Fig. 1).

Synthesis of TiO_2 NPs

After centrifugation of the culture media, the resulting biomass was dried in an oven at 25 °C. 1 g of dry weight was boiled in 10 mL of double-distilled water for 15 min at 100 °C in an Erlenmeyer flask. After boiling, the mixture was cooled and centrifuged at 10,000 rpm for 15 min. The supernatant was collected and stored at 4 °C. Cell-free extracts of the cyanobacterial strains were used for TiO_2 NPs. 40 mL of 25 mM titanium hydroxide solution was added to the mixture and maintained at 37 °C with constant shaking for 24 h. The mixture was then left at room temperature for 8 h; formation of a white precipitate indicated TiO_2 NPs production. The precipitate was separated from the medium by centrifugation at $3820 \times g$ for 15 min. Finally, it was washed with water and subjected to spectrophotometric analyses¹⁶ (Fig. 2).

Preparation of PVA/ TiO_2 NCs

Initially, a 10% (w/w) aqueous solution of PVA was prepared. Various concentrations of TiO_2 NPs (0, 0.5, 1, and 2%) were added to the solution and stirred at room temperature for 60 min to ensure uniformity. 1.5% (w/w) glycerol was added as a plasticizer. Finally, the obtained solution was poured into Petri dishes and kept at room temperature for 48 h for evaporation and film formation. After separating the films from the plates, they were placed in an oven at 50 °C for 5 h to ensure complete drying¹⁷ (Fig. 3).

Characterization of the wound healing cream

The spreadability of the wound healing cream, indicating its ability to be uniformly distributed over the skin surface, was evaluated visually¹⁸. Viscosity was measured at 4 °C using a Brookfield viscometer equipped with



Fig. 1. Cultivation of *N. persica* strain in liquid Z8 medium.

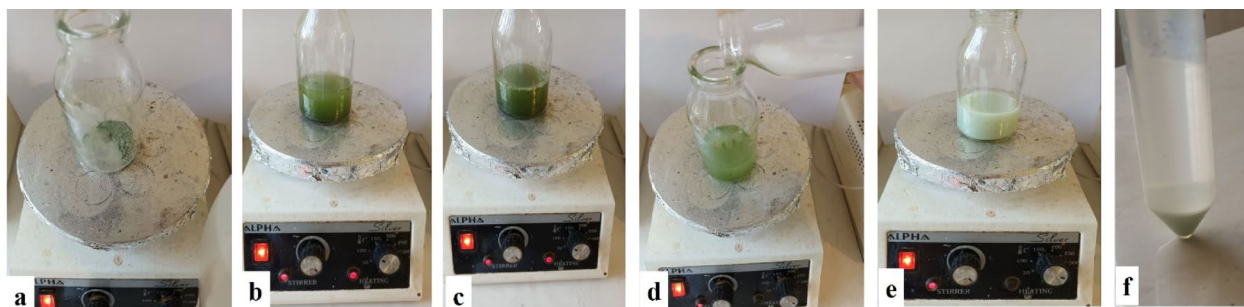


Fig. 2. Synthesis of TiO_2 NPs. (a) Dried biomass, (b) addition of distilled water, (c) boiling, (d) addition of 25 mM titanium hydroxide solution, (e) shaking of the obtained solution, (f) formation of white precipitate indicating TiO_2 NPs production.

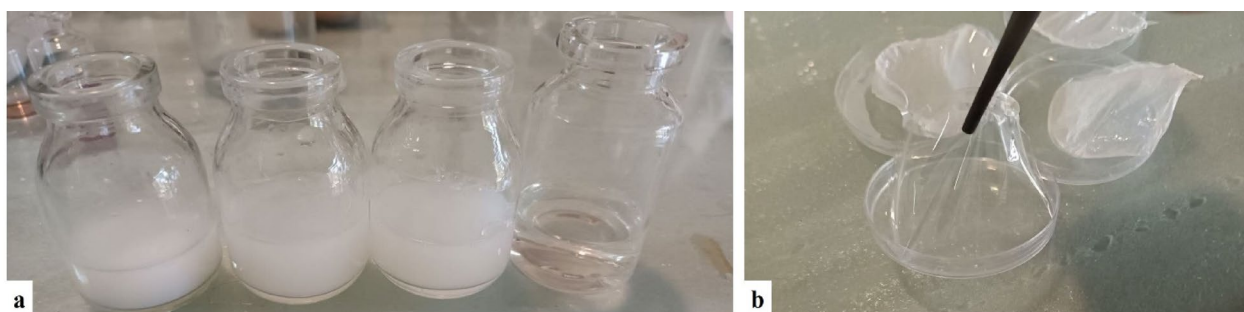


Fig. 3. (a) Preparation of various concentrations of TiO_2 NPs, (b) Preparation of PVA/ TiO_2 NCs.

spindle No. 64 at 100 rpm¹⁹. The pH of the formulations was determined at 25 ± 1 °C using a calibrated pH meter²⁰.

Homogeneity was assessed by visual inspection of the cream after placement in a container, ensuring uniform appearance without lumps or phase separation²¹.

For stability evaluation, the wound healing cream was stored for 4 weeks at 4, 25, 37, and 40 °C. Stability was determined by monitoring changes in color, phase homogeneity, and overall appearance²².

In vivo wound healing cream evaluation

Preparation of wound healing cream

The cream was prepared from liquid and solid phases. The liquid phase consisted of 3 g of coconut oil, 10 g of liquid paraffin, and 5 g of glycerin. The solid phase consisted of 5 g vaseline, 4 g almond oil, 5 g beeswax, 5 g stearic acid, and 3 g glyceryl monostearate. 0.1 g of 2% PVA NPs powder was added per 100 g of cream. The films were pulverized by pounding. The components were combined with 60 mL of water and 2 g of propolis (Fig. 4)²³.

Artificial infection of mice with tested *C. albicans*

Adult mice (strain: C57BL/6), aged 8–12 weeks, were used. Both sexes were included in the experiments. Mice were divided into four groups of three. Each mouse was anesthetized with 100 mg/kg ketamine and 5 mg/kg xylazine injection. The hair on their back skin was shaved using a sterile razor. The incision area was sterilized with 70% ethanol, and then a circular lesion approximately 20 mm in diameter was created on the disinfected skin. Mice were subcutaneously injected with 1 mL of fungal suspension (10^6 CFU/mL) into the wounded area. To confirm infection occurrence, scales were taken from the lesion. They were then cultured on Sabouraud dextrose agar (SDA; Merck, Germany) medium with olive oil, leading to the growth of new *C. albicans* colonies²⁴(Fig. 5).

Treatment of infected wounds with wound healing cream

To test the efficacy of the PVA, TiO_2 NCs obtained from cyanobacteria and propolis as a topical treatment, the prepared creams were applied to the infected and wounded site for 14 days according to the schedule specified in Table 1.

Cream assays

Antifungal activity of wound healing cream by disk diffusion method and determination of minimum inhibitory concentration (MIC) and minimum fungicidal concentration (MFC)

The tested pathogenic fungi, *C. albicans* (PTCC 5199) and *T. rubrum* (PTCC 5143), were obtained from the Iranian Industrial Microorganisms Collection Center. To determine the antifungal activity of the nanoparticle-containing wound healing cream against various human pathogenic fungi and measure the inhibition zone

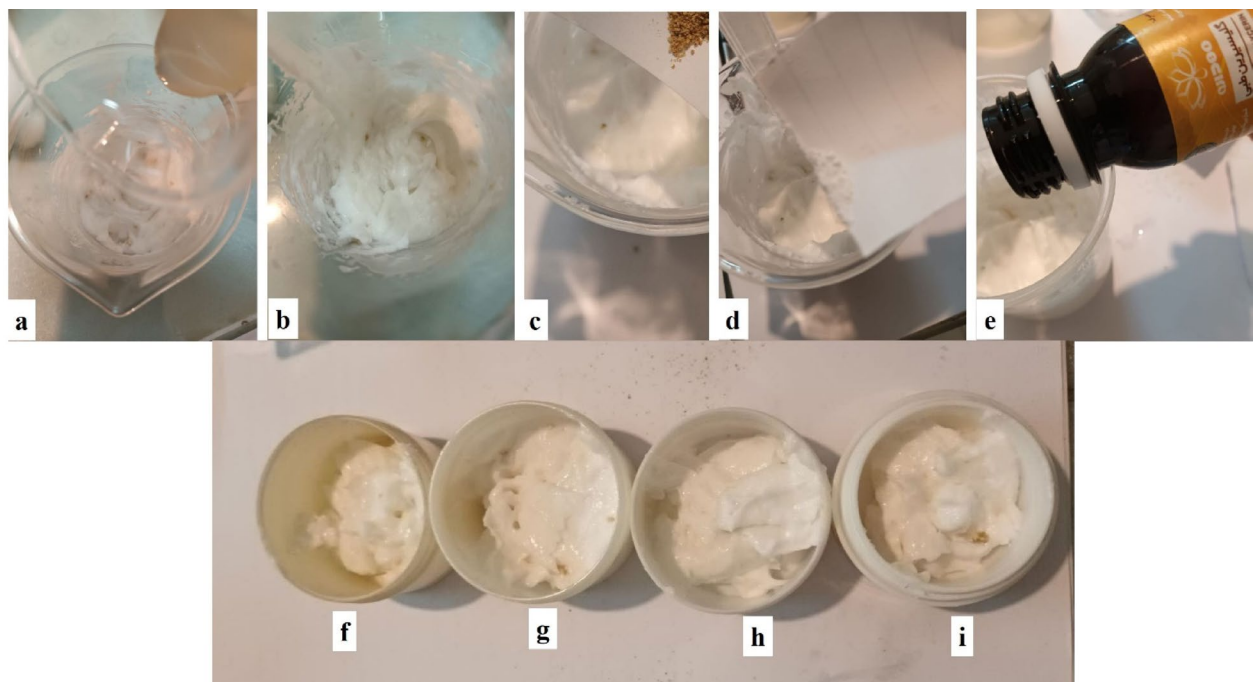


Fig. 4. Preparation of wound healing cream. (a) and (b) mixing of liquid and solid phases, (c) addition of propolis, (d) addition of 2% PVA NPs powder, (e) addition of glycerin, (f) cream containing nanoparticles and propolis, (g) nanoparticle cream, (h) propolis cream, (i) control cream.



Fig. 5. Artificial infection of mice with tested *C. albicans*. (a) anesthetizing the mouse, (b) shaving excess hair, (c) and (d) marking the wound site, (e) and (f) creating the wound, (g) creating fungal contamination, (i) bandaging the infected wound site.

Group	Day	Formulation
C0	Day 1	Inoculation with 1 mL normal saline solution
	Days 3–14	No topical cream application
T1	Day 1	Inoculation with 10 ⁶ CFU/mL <i>C. albicans</i> fungus
	Days 3–14	Topical application of base cream without nanoparticles and propolis (twice daily)
T2	Day 1	Inoculation with 10 ⁶ CFU/mL <i>C. albicans</i> fungus
	Days 3–14	Topical application of propolis cream without nanoparticles (twice daily)
T3	Day 1	Inoculation with 10 ⁶ CFU/mL <i>C. albicans</i> fungus
	Days 3–14	Topical application of nanoparticle cream without propolis (twice daily)
T4	Day 1	Inoculation with 10 ⁶ CFU/mL <i>C. albicans</i> fungus
	Days 3–14	Topical application of cream containing nanoparticles and propolis (twice daily)

Table 1. Research treatment formulations.

diameter, each fungus was first inoculated into 0.5 mL of sterile Sabouraud's Dextrose Agar at approximately 10⁶ cells per mL for *C. albicans* and *T. rubrum*. Before solidification, they were allowed to cool. Each disk was then aseptically filled with 100 µL of each extract at 5% concentration prepared with different solvents. Clotrimazole cream (1% w/w, commercial formulation, Canesten®, Bayer, Germany) was used as the positive control under the same experimental conditions as the tested formulations.

T. rubrum and *C. albicans* plates were incubated at 37 °C for 15 days and 28 °C for 3 days, respectively. After incubation, the inhibition zone diameter was measured²⁴.

This assay was performed for all 4 creams and both fungi. After inoculating *C. albicans* and *T. rubrum* with a 10⁶ cells/mL suspension, MIC was determined by microdilution method. For the experiment, the first group consisted of 9 test tubes; 1 mL of Sabouraud dextrose broth was added to each, and 1 mL of cream solution diluted with Ringer's was added to the first tube. Then, 1 mL from the first tube was transferred to the second, continuing to the ninth tube, and finally 1 mL from the last tube was discarded to equalize volumes. Subsequently, 100 µL of 0.5 McFarland was added to each of the 9 tubes, and the same was done for all 4 groups along with clotrimazole, followed by incubation at 25 °C for 3 to 5 days. Three replicates were used for each culture medium and concentration to minimize experimental error, and the antimicrobial effects of different dilutions were examined²⁵.

Finally, tube concentrations were prepared at 50, 25, 12.5, 6.25, 3.125, 1.56, 0.78, and 0.39 mg/mL. Inoculation of tubes with diluted cream solution was completed in all tubes except the negative control tube, which was inoculated with 100 µL of the extraction solvent. MIC was recorded as the lowest extract concentration preventing pathogenic fungal growth. MFC was recorded as the lowest concentration causing fungal death.

Examination of nanoparticle-containing wound healing cream by transmission electron microscopy (TEM)

For TEM analysis, nanoparticle- and propolis-containing cream samples were collected by centrifugation and treated with 2.5% glutaraldehyde in 0.1 M phosphate buffer (pH 7.4) at 4 °C for 2 h. Then, 1% osmium tetroxide was used as a post-fixative (4 °C, 1.5 h). Samples were dehydrated with graded ethanol series (50, 70, 90, 95, and four times 100%, each for 30 min) followed by acetone dehydration for 30 min. Finally, epoxy resin was used for embedding fixed samples. Ultramicrotome was used for cutting ultra-thin and semi-thin sections. Semi-thin Sect. (1 µm to 850 nm) were stained with 1% toluidine blue and observed under a light microscope using Olympus BX61. Ultra-thin sections of 70 to 90 nm thickness were cut, stained with 2.5% uranyl acetate as primary stain and lead citrate as counterstain. Finally, ultra-thin sections were examined using JEM-1400 Plus TEM²⁶.

DPPH antioxidant activity assay

To evaluate the antioxidant properties of the nanoparticle-containing wound healing cream, the DPPH (2,2-diphenyl-1-picrylhydrazyl) radical scavenging technique was used. Methanol was used to dissolve samples, and methanolic DPPH served as the control. Absorbance of the reaction mixture was measured at 517 nm using a spectrophotometer. Ascorbic acid was used as the control, and the DPPH free radical inhibition percentage was determined by the following equation²⁷:

$$(Ac - As) / Ac \times 100 = \% \text{ inhibition}$$

Where Ac: absorbance of control and As: absorbance of sample.

Cytotoxicity assay

The MTT (3-(4,5-dimethylthiazol-2-yl)-2,5-diphenyltetrazolium bromide) assay was performed on HFFC human foreskin fibroblast cell lines to evaluate the inhibitory effects of materials on cell proliferation. The HFFC cell line was purchased from the National Cell Bank of Iran (NCBI), Pasteur Institute of Iran. Cell lines were incubated in RPMI-1640 medium containing 10% fetal bovine serum. Incubation occurred at 37 °C in a 5% CO₂ incubator with 100 units/mL penicillin and 100 µg/mL streptomycin. Cell lines were seeded in a 96-well plate at a density of 1 × 10⁴ cells per well and incubated at 37 °C with 5% CO₂ for 48 h. After incubation, cells were treated with serial dilutions of four different wound healing cream formulations (control cream, nanoparticle cream,

propolis cream, and nanoparticle–propolis cream) prepared in RPMI-1640 medium. The tested concentrations (0–200 ppm; 0, 0.195, 0.39, 0.78, 1.56, 3.125, 6.25, 12.5, 25, 50, 100, and 200 ppm) represent dilutions of the complete cream formulations (w/v) rather than individual components. After 24 h of treatment, 20 μ L of 5 mg/mL MTT solution was added and incubated for 4 h. To dissolve the produced purple formazan, 100 μ L dimethyl sulfoxide (DMSO) was added to each well. The assay was measured and recorded at 570 nm absorbance using a plate reader²⁸.

Wound healing measurement

The healed wound area was measured in millimeters on days 0, 3, 7, 10, and 12. The wound healing percentage was then calculated using the following equation²⁴.

$$100/20 \text{ (treated area in mm)} = \text{wound healing percentage}$$

Note: No positive control drug was included in the in vivo wound healing model, as clotrimazole was only used as a reference antifungal agent in vitro and is not a standard drug for wound healing studies.

Light microscopy histological examination

For histological evaluation, healthy mouse skin served as the positive control, and infected and treated samples from four experimental groups were examined: group 1, *C. albicans*-infected skin treated with base cream without propolis and nanoparticles (therapeutic negative control); group 2, infected skin treated with propolis-containing cream without nanoparticles; group 3, infected skin treated with nanoparticle-containing cream without propolis; and group 4, infected skin treated with cream containing nanoparticles and propolis. This grouping aimed to compare infected and healthy skin as well as evaluate the efficacy and topical safety of nanoformulations.

On day 14 after infection and treatment initiation, full-thickness skin samples were collected under sterile conditions (Fig. 6) and immediately fixed in 10% buffered formalin for 24–48 h. Routine tissue processing steps followed, including washing, dehydration in graded increasing alcohol concentrations, clearing in xylene, infiltration, and embedding in molten paraffin at approximately 50–56 °C, ultimately forming paraffin blocks. From the blocks, 5 μ m thick sections were prepared using a microtome, transferred to glass slides, and dried²⁴.

For general tissue examination, sections were stained with hematoxylin–eosin (H&E). Slides were first deparaffinized in xylene, then rehydrated through decreasing alcohol concentrations to distilled water. After hematoxylin staining, differentiation in acid alcohol, bluing in alkaline water, and counterstaining with eosin were performed. Finally, sections were rehydrated in increasing alcohols, cleared in xylene, and mounted with resinous mounting medium.

For collagen deposition and dermal matrix regeneration evaluation, Masson's trichrome staining was used. After deparaffinization and rehydration, sections were stained with iron hematoxylin, then acid fuchsin, differentiated in phosphomolybdic-phosphotungstic solution, and stained with aniline blue (or light green). Washing in 1% acetic acid, dehydration in increasing alcohols, clearing in xylene, and mounting were the final steps. In this staining, collagen fibers appeared blue, and cell nuclei dark blue against a red background. Histopathological examinations were performed using BX51 light microscope (Olympus) at 400 $\times$ magnification. For each animal, several fields of view from each section were evaluated blindly by a pathologist unaware of treatment groups. Measured indices included numbers of mono- and multinucleated inflammatory cells, fibrocytes and fibroblasts, degree of epithelialization, and collagen fiber organization. Epithelialization on day 14 was semi-quantitatively scored on a five-point scale: 0 (no epithelialization), 1 (25%), 2 (50%), 3 (75%), and 4 (100% complete epithelial coverage). Cell counting was performed at 400 $\times$ magnification²⁴.

Data analysis methods and tools

All experiments were conducted in triplicate, and results reported as mean $\pm$ standard deviation. Experimental data were compared using one-way analysis of variance (One-way ANOVA), and significant statistical differences between means (when overall treatment effect was significant) determined using Duncan's multiple range test. Statistical test results were obtained using SPSS version 26 software.

Results

TiO₂ nanoparticles spectroscopic analysis

Spectrometry results of TiO₂ NPs showed absorption of approximately 0.614 at λ max 382 nm. This result indicates that TiO₂ NPs were successfully prepared (Fig. 7).

Characterization of the wound healing cream

The spreadability of the wound healing cream was found to be satisfactory throughout the 30-day storage period, indicating good applicability and consistency of the formulations.

The viscosity of all formulations showed a slight increase over the 8-week storage period. Overall, no significant phase instability or abrupt changes were observed in any formulation. The results indicate that incorporation of different concentrations of TiO₂ nanoparticles had no adverse effect on the rheological stability of the cream.

The pH values of all formulations remained close to neutral throughout the 30-day period, ranging approximately from 7.07 to 6.97. A slight gradual decrease in pH was observed over time; however, all formulations remained within an acceptable range for topical application. No significant destabilizing effect of TiO₂ nanoparticle concentration was observed.

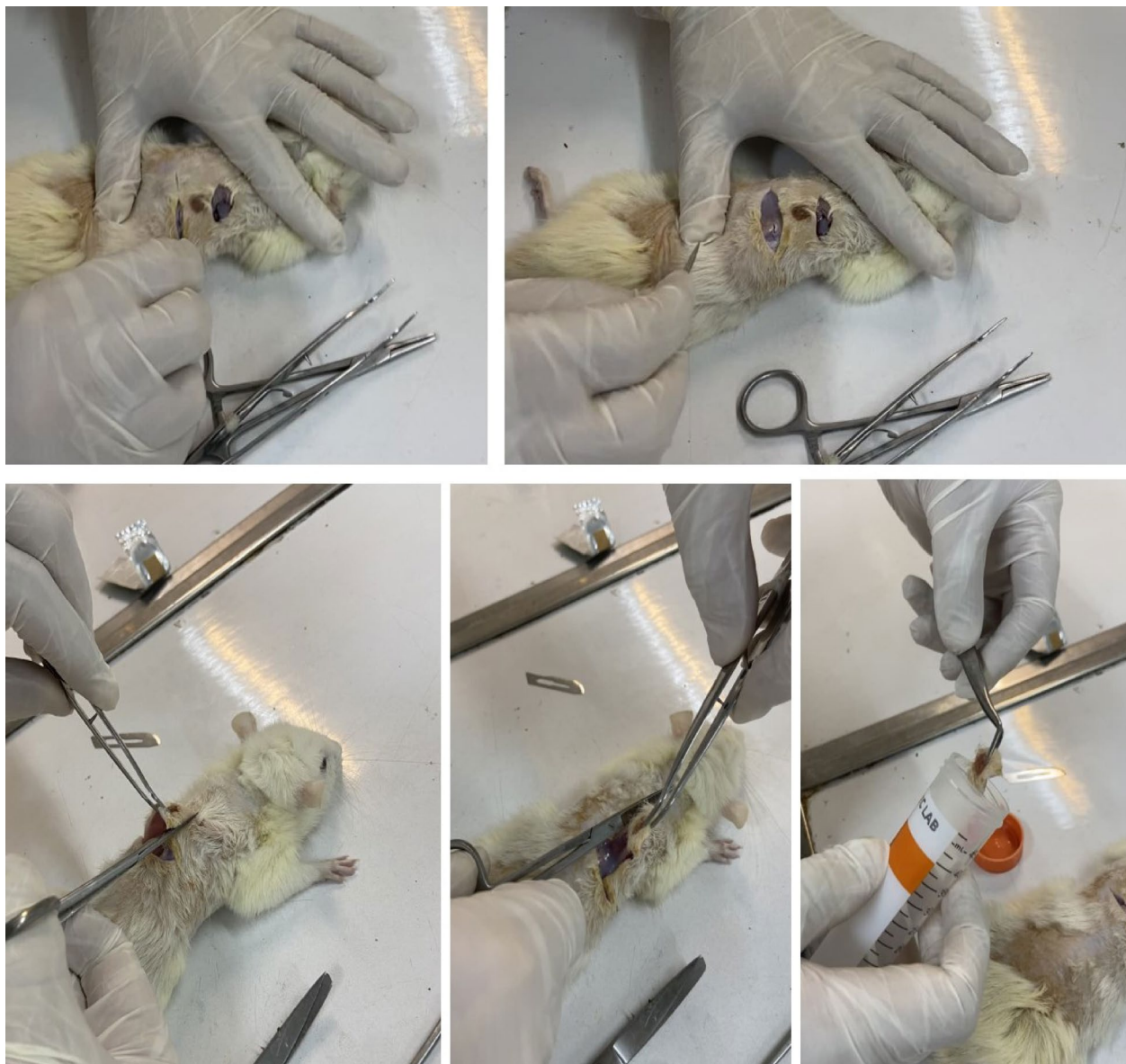


Fig. 6. Isolation of skin tissue from mice treated with prepared creams after 14 days.

The homogeneity results indicated that incorporation of different concentrations of TiO_2 nanoparticles did not affect the physical uniformity of the wound healing cream. All formulations remained homogeneous throughout the 30-day storage period, with no visible phase separation, aggregation, or lump formation observed.

The color stability results demonstrated that formulations stored at 25 °C showed minimal changes over time, whereas storage at elevated temperatures (37 and 40 °C) led to gradual and concentration-dependent changes in color intensity. The most pronounced changes were observed at 40 °C after 30 days, particularly in formulations with lower TiO_2 concentrations, indicating a temperature-dependent decrease in color stability.

Despite these changes, formulations containing higher TiO_2 concentrations showed better color stability under stress conditions.

Antifungal activity of the wound healing cream

Statistical analysis using one-way ANOVA showed significant differences in inhibition zone diameters against *C. albicans* (PTCC 5199) and *T. rubrum* (PTCC 5143), with the positive control treatment (clotrimazole) exhibiting the highest inhibition zone diameter with significant difference in both cases. Among treatments against both fungi, the cream containing nanoparticles and propolis showed the highest inhibition zone diameter with significant difference ($p < 0.05$). Additionally, the aforementioned treatment exhibited the highest inhibition zone diameter against *T. rubrum* with significant difference. The lowest inhibition zone diameter against both fungi was reported for the control treatment (Figs. 8 and 9, and 10).

MIC calculations revealed that the lowest inhibitory concentrations against both fungi belonged to creams containing nanoparticles and propolis, propolis cream, and nanoparticle cream, with no significant difference

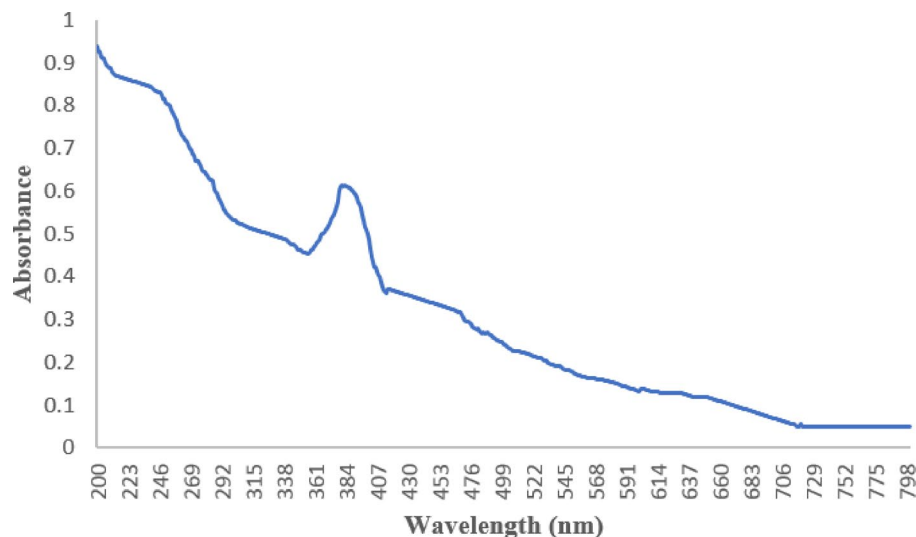


Fig. 7. Results from spectroscopy of TiO₂ NPs.

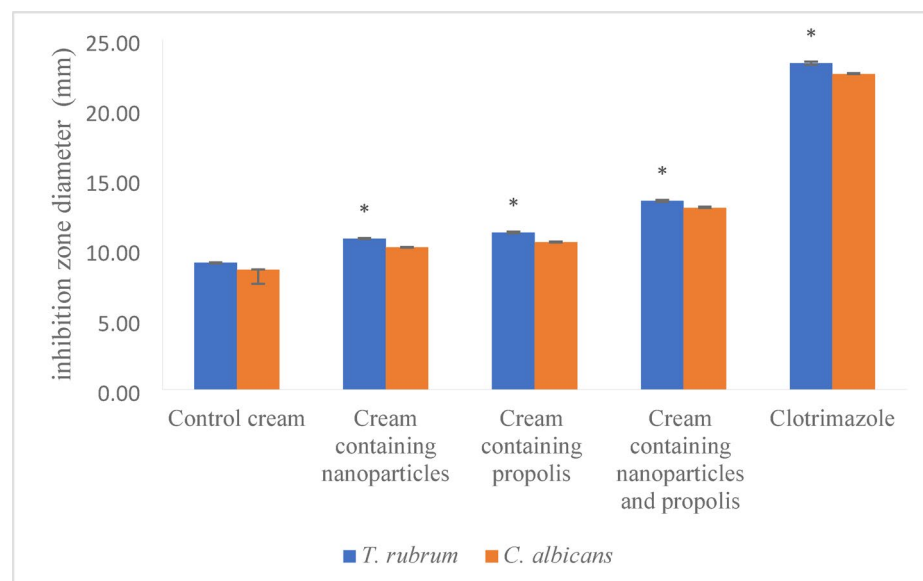


Fig. 8. Results of comparing inhibition zone diameters of prepared creams with different treatments and positive control against *C. albicans* and *T. rubrum*. Statistical significance was determined using $p < 0.05$. * indicates a significant difference compared to the control group.

between the aforementioned creams. Comparison of antifungal activity among the tested creams showed the lowest MIC against *T. rubrum* with significant difference ($p < 0.05$) (Fig. 11).

MFC calculations indicated that the lowest inhibitory concentrations against both fungi belonged to creams containing nanoparticles and propolis, propolis cream, and nanoparticle cream, with no significant difference between the aforementioned creams. Comparison of antifungal activity among the tested creams showed the lowest MFC against *T. rubrum* with significant difference. Additionally, the control cream showed no MFC activity ($p < 0.05$) (Fig. 12).

TEM characterization of the nanoparticle-containing wound healing cream

Results of inhibition zone diameters, MIC, and MFC for the cream containing nanoparticles and propolis showed that *C. albicans* is much more resistant than *T. rubrum*. Therefore, subsequent analyses focused on the effects of prepared creams on the resistant fungus *C. albicans*. As shown in Fig. 13, micrographs of this sample reveal particles with quasi-spherical morphology. To examine particle size distribution in the sample, multiple particles from TEM images were measured using ImageJ image processing software, and the size distribution histogram is shown in Fig. 14. In the histogram of Fig. 14, the horizontal axis represents particle sizes observed

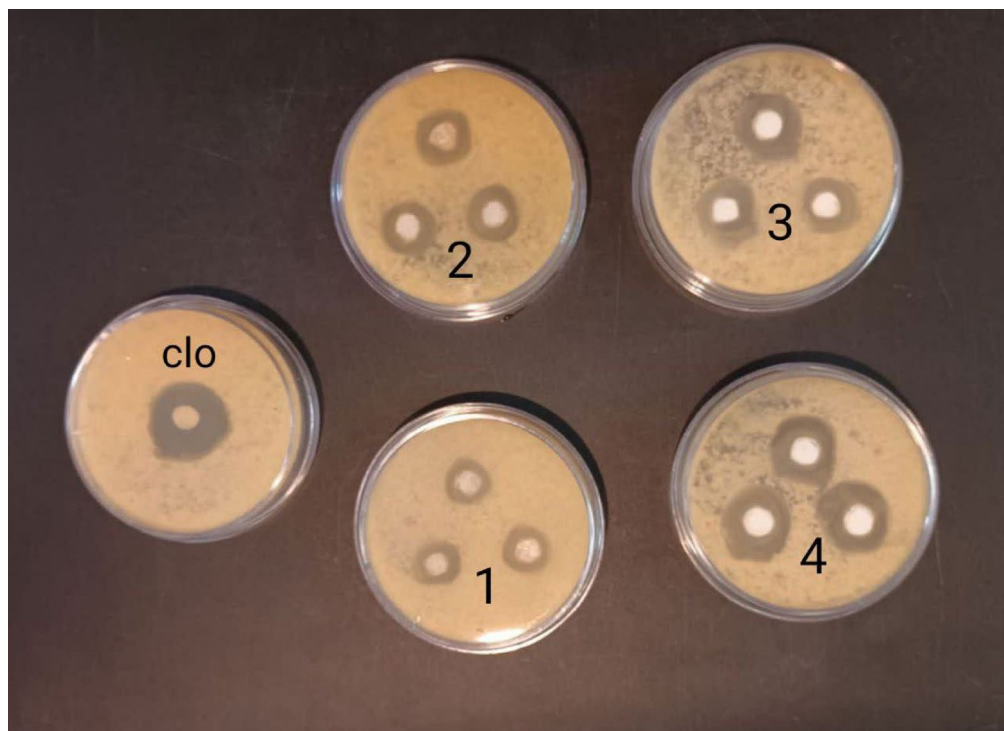


Fig. 9. Results of inhibition zone diameters of prepared creams with different treatments and positive control against *C. albicans*. T1: cream without nanoparticles and propolis, T2: propolis cream without nanoparticles, T3: nanoparticle cream without propolis, T4: cream containing nanoparticles and propolis.

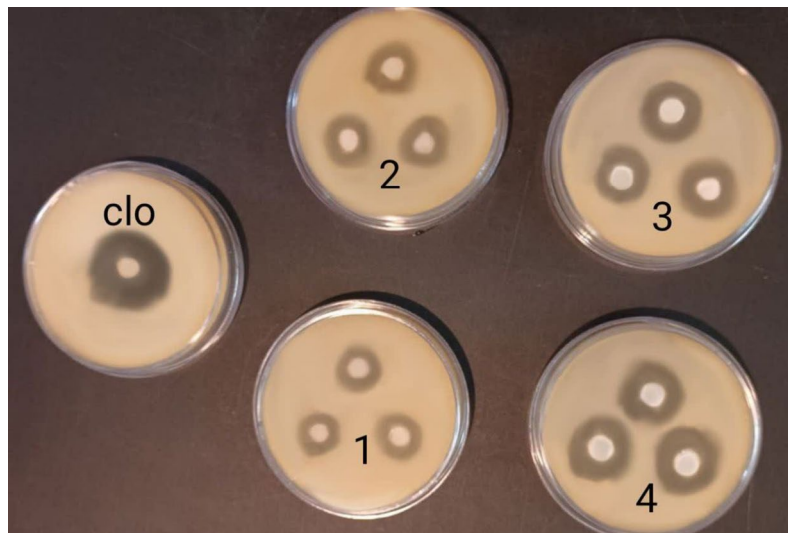


Fig. 10. Results of inhibition zone diameters of prepared creams with different treatments and positive control against *T. rubrum*. T1: cream without nanoparticles and propolis, T2: propolis cream without nanoparticles, T3: nanoparticle cream without propolis, T4: cream containing nanoparticles and propolis.

in the TEM micrograph, and the vertical axis represents the number of particles measured in that size range by the software. According to the histogram shown in Fig. 14, all particles in the sample were in the range of 20 to 100 nm, with most particles in the 40 to 50 nm range. Statistical data from these measurements are reported in Table 2. According to Table 2, the mean particle size in the sample was 57.03 ± 17.63 nm. Based on these results, this particle size distribution can be considered relatively broad and fairly normal. The smallest, largest, and median particle sizes measured are also reported in Table 2 as 27.15, 52.41, and 95.75 nm, respectively. The closeness of the median value to the mean in these statistical results provides further evidence of the normal distribution of particle sizes in this sample.

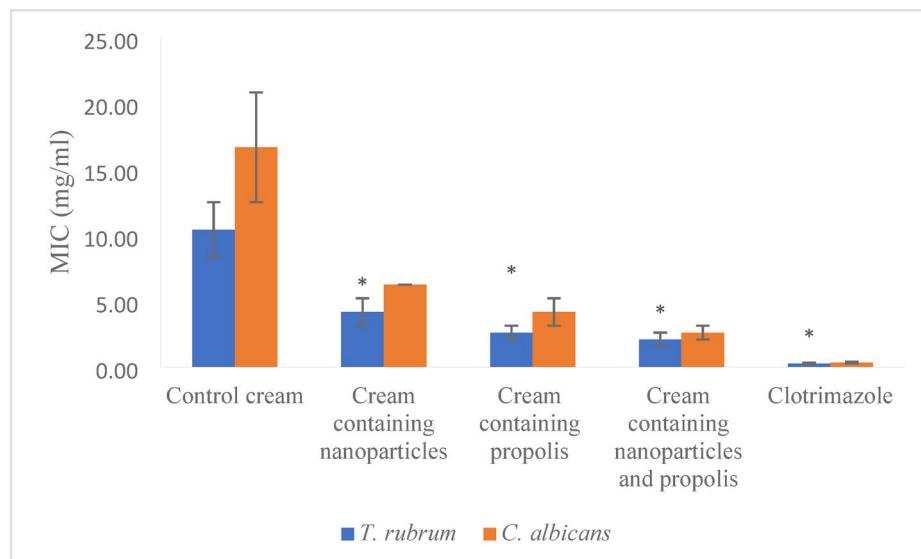


Fig. 11. Results of comparing MIC measurements against *C. albicans* and *T. rubrum*. Statistical significance was determined using $p < 0.05$. * indicates a significant difference compared to the control group.

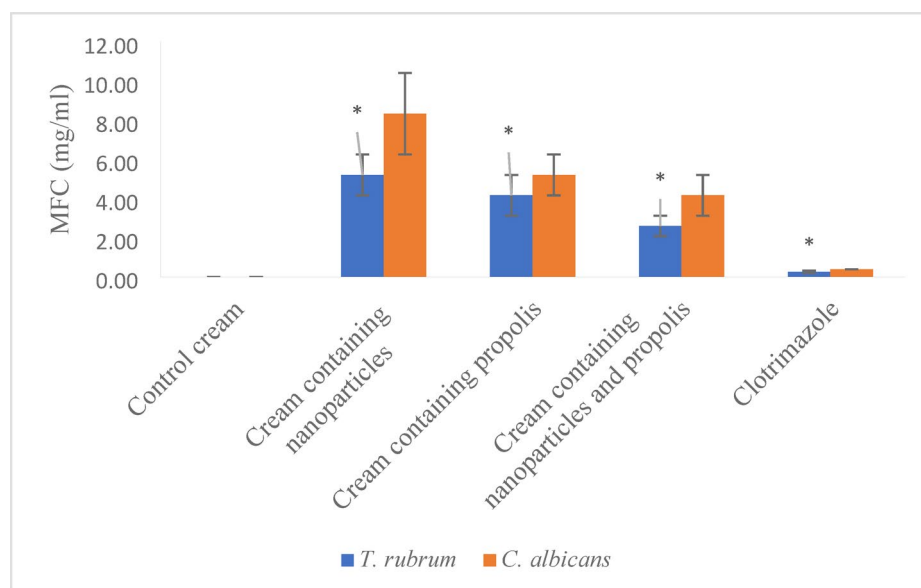


Fig. 12. Results of comparing MFC measurements against *C. albicans* and *T. rubrum*. Statistical significance was determined using $p < 0.05$. * indicates a significant difference compared to the control group.

TEM analysis of *Candida albicans* interaction with the nanoparticle-containing cream

TEM images clearly show close contact and gradual destruction of *C. albicans* spore structure in the presence of nanoparticles (Fig. 15). In all four images, darker particles are visible around and on the spore wall, indicating nanoparticle accumulation and initiation of penetration into the cell structure. (a) Spore with relatively complete and thick wall is visible, but wall thickness is uneven in some areas with localized breakage or thinning observed. Nanoparticle accumulation appears as dark clumps on the surface and around the wall, causing delamination and cytoplasmic content leakage at several points; this is consistent with cell lysis onset and membrane integrity loss. (b) Spore appears almost as a ring with relatively uniform wall, but the wall edge is severely serrated and irregular, indicating wall destruction and erosion by nanoparticles. Nanoparticle accumulation forms a dark necklace or ring completely surrounding the spore, meaning complete cell encirclement and creation of a nanoparticle layer in direct contact with the wall; this sets the stage for subsequent internal penetration. (c) In this view, the spore has an oval shape with multi-layered and irregular wall; wall layers are separated at some points with rough edges, indicating wall swelling and destruction. Nanoparticles appear as dark spots and clusters at the wall boundary and internal cell space, indicating gradual nanoparticle penetration into the spore

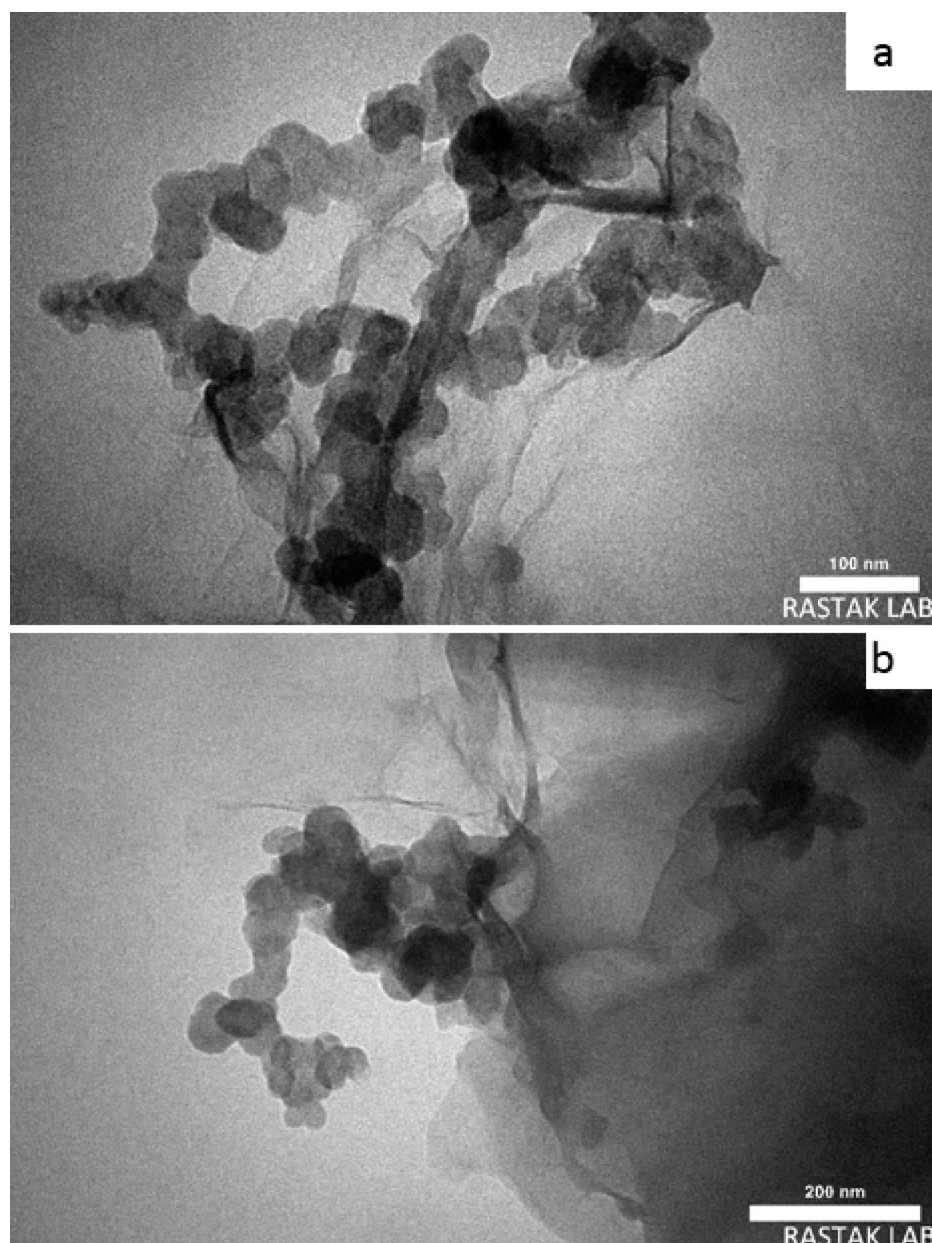


Fig. 13. TEM micrographs of cream containing nanoparticles and propolis at 100 nm (a) and 200 nm (b) magnification.

and disruption of internal organization (cytoplasm and organelles). (d) Spore appears circular with relatively light center, but the wall is completely irregular and thickened with clear protrusions and depressions in some areas, indicating advanced wall destruction. A dense, very dark nanoparticle ring has formed around the spore, with dark clumps extending inward at several points; this pattern represents the stage where nanoparticles, besides completely surrounding the spore, have penetrated the internal cell space and likely caused content leakage and cell death (Fig. 15).

DPPH radical scavenging activity

The antioxidant activity of the formulations was evaluated using the DPPH assay based on IC_{50} values (Fig. 16). The control cream exhibited the highest IC_{50} value, indicating the lowest antioxidant activity, while formulations containing propolis showed significantly improved antioxidant capacity. Both the propolis-containing cream and the nanoparticle–propolis formulation demonstrated significantly lower IC_{50} values compared to the control cream and the nanoparticle-only cream ($p < 0.05$). However, no significant difference was observed between the propolis-only cream and the nanoparticle–propolis formulation, indicating that the addition of nanoparticles did not further enhance antioxidant activity.

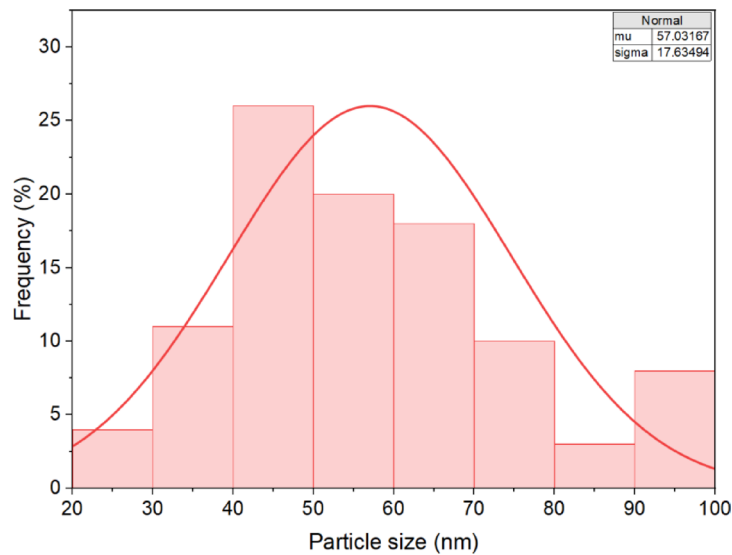


Fig. 14. Particle size distribution histogram of the examined sample.

Total measurements	Mean (nm)	Standard deviation (nm)	Smallest measured particle (nm)	Median measured particle (nm)	Largest measured particle (nm)
100	57.03	17.63	27.15	52.41	95.75

Table 2. Statistical results from histograms shown in examined samples.

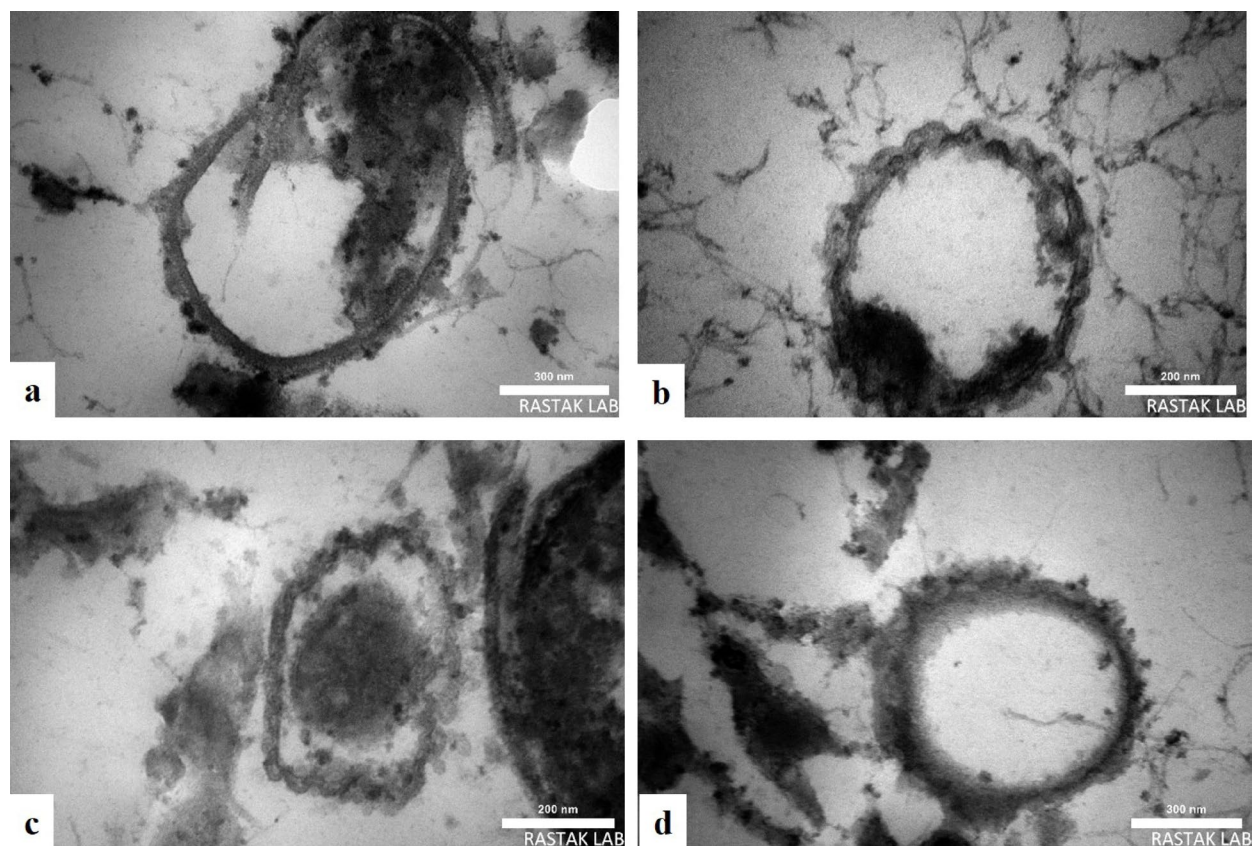


Fig. 15. Nanoparticle attack on *C. albicans* fungal spores, encircling the wall and penetrating into the cell.

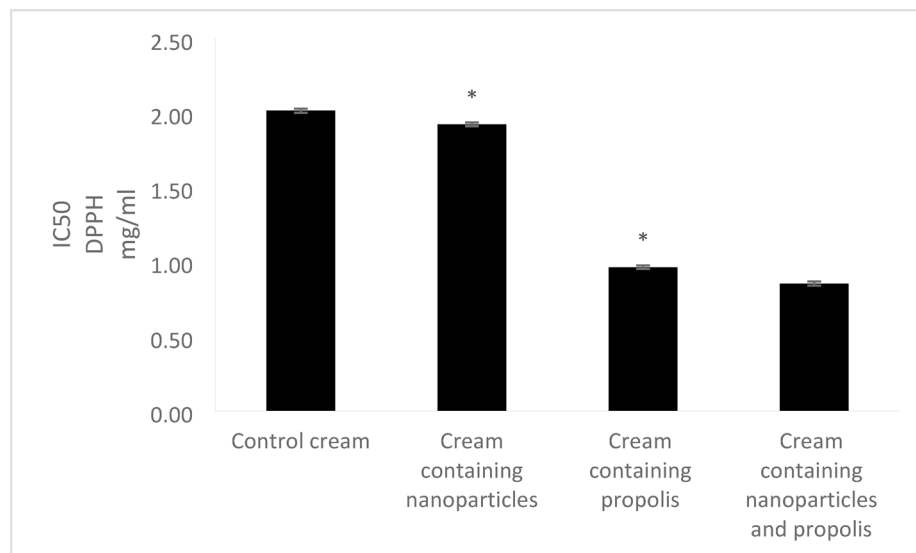


Fig. 16. Calculation of IC₅₀ values and statistical differences in the antioxidant activity of prepared creams. Statistical significance was determined using $p < 0.05$. * indicates a significant difference compared to the control group.

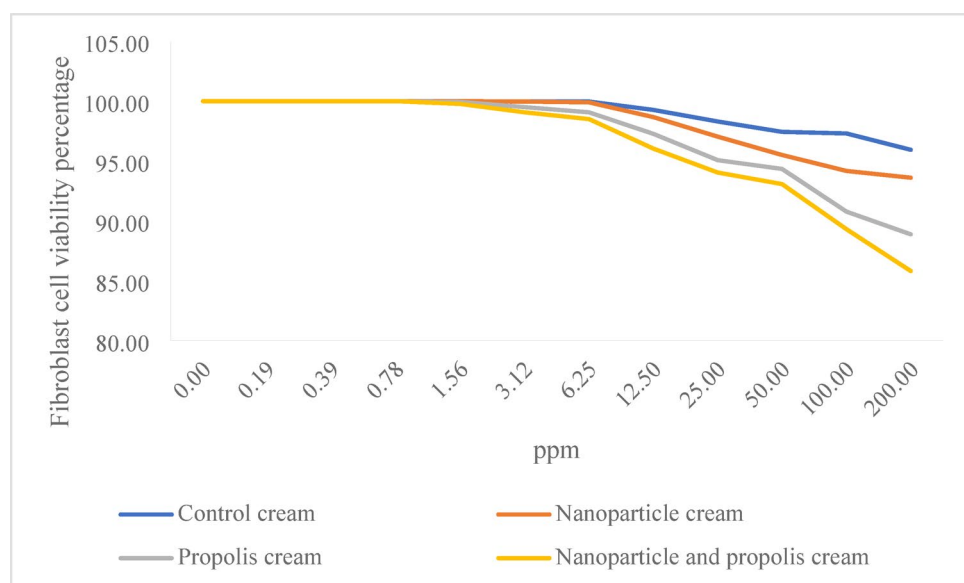


Fig. 17. Fibroblast cell viability percentage against prepared creams with different treatments.

Cytotoxicity (MTT) assay

MTT assay results showed that fibroblast cell viability decreased slightly with increasing concentrations (0–200 ppm) of all tested wound healing cream formulations (control cream, nanoparticle cream, propolis cream, and nanoparticle–propolis cream). All tested concentrations represent serial dilutions of the complete cream formulations in culture medium. Among the tested groups, the nanoparticle–propolis cream exhibited a slightly higher reduction in cell viability compared to the other formulations, reaching $85.80 \pm 0.26\%$ viability at 200 ppm. At low concentrations (≤ 0.78 ppm), both the propolis-containing cream and the nanoparticle–propolis cream showed no significant cytotoxic effect on fibroblast cells, with viability remaining comparable to the control group. In contrast, for the control cream and nanoparticle-only cream, no significant cytotoxic effect was observed up to 1.56 ppm. Statistical analysis indicated that differences at low concentrations were not significant ($p > 0.05$), while higher concentrations showed statistically significant variations among treatment groups ($p < 0.05$), as shown in Fig. 17.

Wound healing assessment

Wound healing assessment showed that statistical analysis of wound closure (healed area reduction) on days 0, 3, 7, 10, and 14 demonstrated a progressive increase in healing over time. However, creams containing propolis and nanoparticles exhibited significantly greater wound closure rates compared to the control cream. Importantly, on day 14, no significant difference was observed between the nanoparticle cream and propolis cream, while both showed significant differences compared to the control cream (Fig. 18; Table 3).

No positive control drug was included in the in vivo wound healing model; therefore, comparisons were performed only among the prepared cream formulations.

Macroscopic evaluation of infected wounds treated with the wound healing cream

Photographs of the healed wound area on days 0, 3, 7, 10, and 14 in all four examined mouse groups clearly show significant wound healing over 14 days.

Histopathological examination of tissue sections

Results of sample evaluation at the 14-day time interval in different groups are shown in Table 4. Histopathological study of skin samples at this time interval was comparatively examined in different groups (Figs. 19 and 20). Healthy skin tissue sample as positive control. Group 1: Skin tissue infected with *C. albicans* treated with cream without nanoparticles and propolis (as untreated negative control). Group 2: Skin tissue treated with propolis

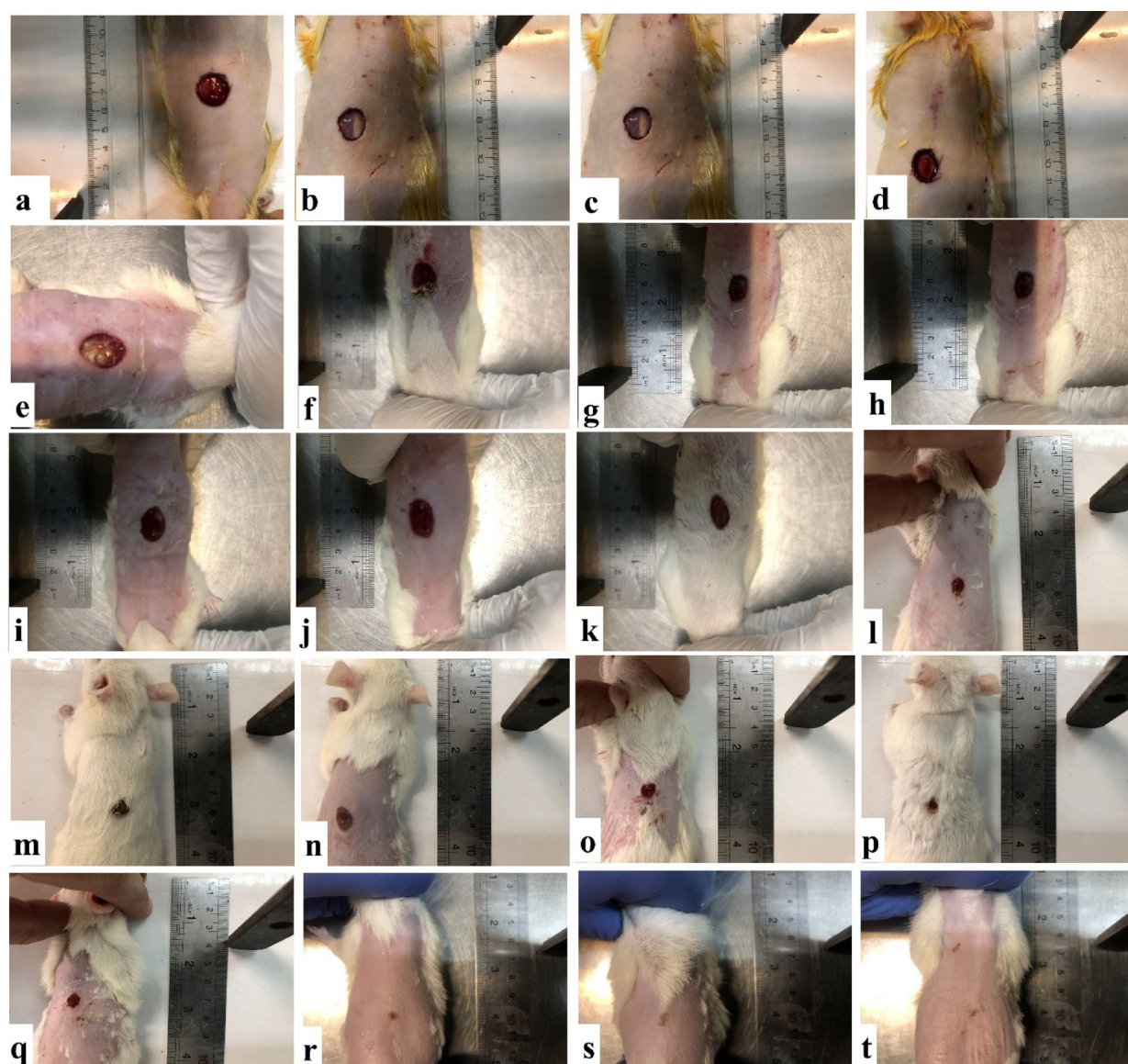


Fig. 18. Photographs of wound creation area in mice treated with control cream, propolis cream, nanoparticle cream, and nanoparticle-propolis cream on day 0 respectively (a), (b), (c), (d); day 3 respectively (e), (f), (g), (h); day 7 respectively (i), (j), (k), (l); day 10 respectively (m), (n), (o), (p); day 14 respectively (q), (r), (s), (t).

	Control cream	Nanoparticle cream	Propolis cream	Nanoparticle and propolis cream
3	4.95 ± 7.02(aA)	28.51 ± 4.18(bA)	31.00 ± 3.07(bA)	40.26 ± 8.72(cA)
7	24.41 ± 2.93(aB)	27.59 ± 10.95(bA)	36.48 ± 7.53(bB)	67.54 ± 15.76(cB)
10	57.98 ± 4.24(aC)	75.06 ± 4.57(bB)	75.57 ± 1.18(bC)	87.92 ± 2.21(cC)
14	83.90 ± 5.06(aD)	97.09 ± 1.88(bC)	98.22 ± 1.78(bD)	99.12 ± 0.88(bD)

Table 3. Statistical calculation results of healed wound area on days 0, 3, 7, 10, and 14 in millimeters. Different lowercase letters indicate significant differences in rows. Different uppercase letters indicate significant differences in columns.

Groups	Group 1	Group 2	Group 3	Group 4
Epithelialization Evaluation				
Epithelium formation	25%	25%	75%	100%
Histomorphometry Evaluation				
Neutrophils	6	5	2	2
Macrophages	3	2	1	1
Collagen Formation Percentage Evaluation				
Collagen formation %	42	70	71	81

Table 4. Evaluation of epithelialization, histomorphometry, and collagen formation percentage in skin samples on day 14 post-treatment.

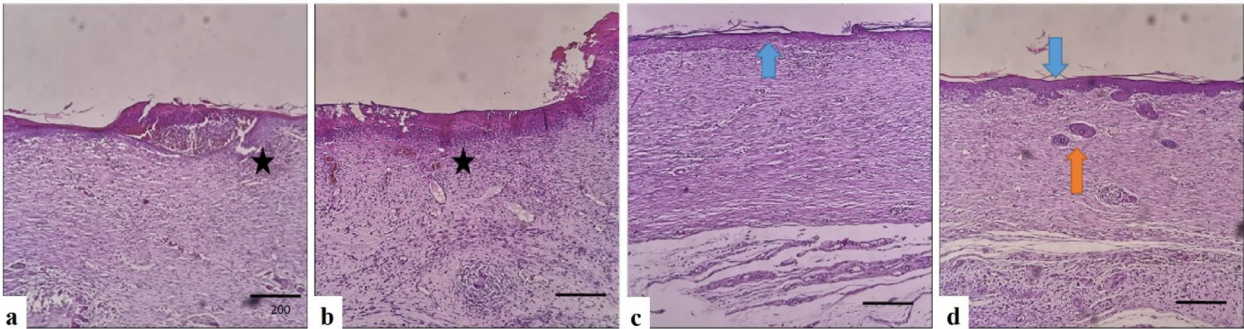


Fig. 19. Hematoxylin-eosin staining at 100× magnification. (Inflammatory cells: star, epithelialization: blue arrow, skin appendages: orange arrow). Group 1, *C. albicans* -infected skin treated with base cream without propolis and nanoparticles (therapeutic negative control); Group 2, infected skin treated with propolis-containing cream without nanoparticles; Group 3, infected skin treated with nanoparticle-containing cream without propolis; and Group 4, infected skin treated with cream containing nanoparticles and propolis.

cream without nanoparticles. Group 3: Skin tissue treated with nanoparticle cream without propolis. Group 4: Mouse skin tissue treated with nanoparticle and propolis cream.

In group 1, very large scabs were observed on the resulting wounds. Additionally, very severe inflammatory reactions were observed in the hypodermis layer, including massive neutrophil infiltration in this area. Findings showed that the epidermal tissue regeneration process had not started, and some acanthotic cells were still present at the wound site. In the second ointment group, inflammatory cell infiltration similar to the first control group was evident. No findings indicating new epithelium formation were observed, and inflammatory cells continued to be seen in the tissue. Histopathological study from group 3 showed that on day 14 after treatment, inflammation in the defect area was somewhat resolved, and a very thin epidermal layer had formed.

In the fourth control group after treatment, the formed epidermal layer had appropriate rete ridges and was formed to varying degrees over the defect. Additionally, no signs of inflammation occurrence were observed in the defect area, and findings showed regrowth of hair follicles and sebaceous glands at the lesion site. Reduction of neutrophils and macrophages after 14 days of wound healing was reported in the examined groups (Table 4).

Wounds in all groups treated with wound healing cream showed more cellular granulation tissue, characterized by spindle and oval cells arranged in long parallel fascicles. Persistent chronic inflammatory infiltration varied from moderate to severe, but collagen fiber content filling intercellular spaces increased compared to the control group. However, edematous changes and inflammation in group 4 were less than in other groups (Table 4).

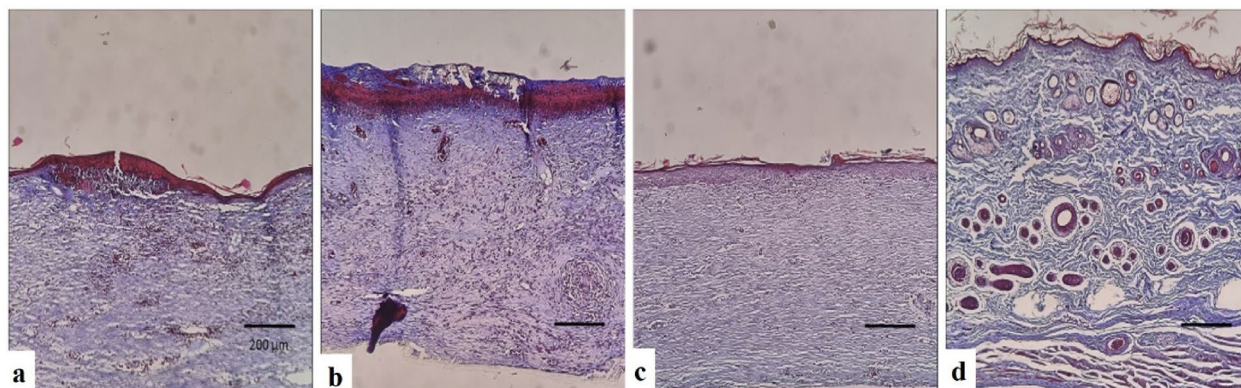


Fig. 20. Masson's trichrome staining at 100× magnification. Group 1, *C. albicans* -infected skin treated with base cream without propolis and nanoparticles (therapeutic negative control); Group 2, infected skin treated with propolis-containing cream without nanoparticles; Group 3, infected skin treated with nanoparticle-containing cream without propolis; and Group 4, infected skin treated with cream containing nanoparticles and propolis.

Discussion

In this study, the use of TiO₂ NPs synthesized by the cyanobacterium *N. persica* in combination with propolis provided a novel approach based on natural biomolecules for designing a cream to treat fungus-infected wounds. The presence of secondary metabolites, polysaccharides, and pigments from cyanobacteria leads to better biocompatibility and colloidal stability, and contributes to maintaining nanoparticles in the nanometric range and preventing particle aggregation, as observed in TEM results and size distribution²⁹. These features provide a suitable platform for topical application of nanostructures on skin, especially when combined with another natural substance like propolis, which itself contains antioxidant and anti-inflammatory polyphenols and flavonoids. One of the main motivations for developing such formulations is the global challenge of drug resistance against antibacterial and antifungal agents that has limited the efficacy of conventional treatments and highlighted the need for alternative options with multiple mechanisms³⁰. Metal and oxide nanostructures, including silver, gold, ZnO, and Fe₃O₄ nanoparticles, have shown a wide range of antimicrobial and antifungal effects in previous studies through cell wall and membrane disruption, penetration into cytoplasm, and induction of reactive oxygen species production³¹.

The TEM data of this research also showed a similar pattern for cyanobacterial TiO₂ nanoparticle adhesion to the *C. albicans* spore wall, creation of irregularities and breakage in the wall, cytoplasmic membrane separation, cytoplasmic vacuolization, and ultimately partial cell lysis were observed. Such changes are consistent with other reports on fungal and bacterial cell structure disruption in the presence of extracts or biogenic nanoparticles²⁴. These observations suggest that nanoparticle interaction with the fungal cell wall may contribute to structural damage and loss of membrane integrity.

The results of antifungal assays in this study demonstrated that the nanoparticle–propolis cream (T4) produced the largest inhibition zone diameter and the lowest MIC and MFC values against *C. albicans* and *T. rubrum*, whereas the base cream (control) showed no inhibitory effect. These findings are consistent with previous reports highlighting the anti-Candidal activity of cyanobacterial extracts and pigments, such as phycocyanin and methanolic *Arthrospira platensis* extract, which have shown significant inhibition zones and low MIC values against similar fungal strains²⁴.

The superior performance of the T4 formulation compared to creams containing only nanoparticles or only propolis suggests an enhanced antifungal effect associated with the combined presence of cyanobacterial-derived TiO₂ nanoparticles and phenolic compounds present in propolis. This improved activity may be attributed to the ability of nanoparticles to disrupt fungal cell wall and membrane integrity, thereby increasing cellular permeability, while propolis compounds interfere with intracellular metabolic pathways and oxidative defense systems. Similar combined or additive effects between metal nanoparticles and bioactive plant or microalgal compounds have been reported to enhance antifungal efficacy³².

Clotrimazole was used as a positive control in this study due to its well-established clinical efficacy as a broad-spectrum antifungal agent against dermatophytes and yeasts, including *C. albicans* and *T. rubrum*. As expected, clotrimazole exhibited stronger antifungal activity compared to the tested formulations. However, despite this difference, the developed nanoparticle–propolis cream offers several important advantages. Unlike conventional antifungal drugs, this formulation combines antifungal, antioxidant, and wound healing properties within a single system. In addition, the use of natural compounds such as propolis may reduce the risk of adverse side effects associated with prolonged use of synthetic antifungal agents. Therefore, this multifunctional formulation may serve as a complementary or alternative therapeutic option, particularly in the management of infected wounds where both antimicrobial activity and tissue regeneration are required.

From an antioxidant perspective, the nanoparticle–propolis cream showed the lowest IC₅₀ and thus the highest radical scavenging power in the DPPH assay, indicating significant enhancement of the

formulation's antioxidant capacity in the presence of both components. Previous studies have shown that cyanobacterial and microalgal extracts, especially species like *Nostoc* and *Arthrospira*, exhibit strong antioxidant activity and metal chelation ability due to high content of polyphenols, flavonoids, carotenoids, and exopolysaccharides³³. Adding propolis, which is rich in flavonoids and phenolic acids, logically can lead to strengthening this capacity, and the results of this study confirm this hypothesis. Such a combination of antioxidant effects is important for wound healing, as reducing oxidative stress in the wound bed helps keratinocyte migration, fibroblast proliferation, and regulation of inflammatory response; the same process reported in studies on phycocyanin-containing creams and herbal healing creams²⁴.

MTT assay results showed that with increasing cream concentration up to 200 ppm, fibroblast cell line viability percentage decreases in a dose-dependent manner, but no significant reduction was observed below about 0.78 ppm in formulations containing propolis and nanoparticle-propolis, and also up to 1.56 ppm in base cream and cream containing only TiO₂ nanoparticles. These findings indicate that the cream demonstrates acceptable biocompatibility within the dose range relevant to topical application, without serious cellular toxicity concerns. At higher concentrations, a reduction in fibroblast viability was observed, particularly in the combined formulation, which may be related to nanoparticle-induced oxidative stress and cellular interactions reported for similar nanostructures³⁴.

Although this study's nanoparticles are TiO₂ not silver, general ROS-based mechanisms and direct cellular structure interactions reported for other nanoparticles can partly explain the observed proliferation reduction³⁵.

Clinical-macroscopic and histopathological wound assessments complemented laboratory findings. Wound area measurements showed significant reduction across all groups over 14 days, but healing speed and percentage closure in the nanoparticle-propolis group were significantly higher than control, nearing complete healing by day 14 while substantial wound area remained open in the base cream. This trend matches studies using phycocyanin-containing or plant/algal extract creams to accelerate wound healing; those researches also showed faster wound reduction and higher healing percentages in algal extract or herbal cream recipient groups versus controls³⁶. Day-14 H&E and Masson's trichrome staining results provided clear tissue evidence of nanoparticle-propolis formulation superiority. In the control group and to some extent in groups containing only propolis or only nanoparticles, thick scabs, severe neutrophil and macrophage infiltration in the hypodermis, incomplete epithelium formation, and collagen disorganization were still observed; findings indicating prolongation of the inflammation phase and delay in entering the reconstruction phase. In contrast, in group 4, inflammation was noticeably reduced, epidermis formed with regular rete ridges and higher epithelialization score, hair follicles and sebaceous glands reappeared, and collagen fiber density and order increased. This pattern is similar to reported results for phycocyanin-containing creams and other algal and plant extracts, where reduced inflammatory cells, better collagen formation, and return of skin appendages were reported as optimal healing indicators²⁴. Therefore, it can be concluded that the cyanobacterial TiO₂-propolis combination not only inhibits fungal growth but also enhances wound healing quality by reducing inflammation, improving oxidative balance, and stimulating tissue regeneration.

Overall, this research shows that the cream containing TiO₂ nanoparticles from cyanobacterium *N. persica* combined with propolis is a multifunctional formulation with strong antifungal properties, high antioxidant capacity, acceptable cellular toxicity at therapeutic doses, and clear ability to accelerate and improve the quality of *C. albicans*-infected wound healing. Alignment of the obtained results with previous reports on antifungal, antioxidant, and healing activities of cyanobacterial and algal compounds strengthens this formulation's position among novel natural resource-based options for managing infected wounds³⁷. However, for translating this product from laboratory to clinical application, further studies including long-term skin safety tests, more detailed examination of molecular mechanisms at gene and protein levels, and ultimately human clinical trials are necessary. A limitation of the present study is the absence of additional physicochemical characterization techniques such as dynamic light scattering (DLS) and zeta potential analysis. Future studies should include these analyses to further elucidate nanoparticle stability and surface charge properties.

Conclusion

This study demonstrated that the nanocream containing TiO₂ nanoparticles derived from cyanobacteria combined with propolis is a promising formulation for the treatment of fungus-infected wounds. TEM analysis showed that nanoparticles exhibited quasi-spherical morphology with a size distribution in the range of ~20–100 nm, with most particles in the 40–50 nm range. These nanoparticles were observed to interact with *C. albicans* cell walls and may contribute to structural damage and membrane disruption. Antifungal assay results showed that the nanoparticle-propolis cream had the highest inhibition zone diameter and lowest MIC and MFC against both *C. albicans* and *T. rubrum*, while the control cream showed no significant antifungal activity. Antioxidant activity (DPPH) indicated that the nanoparticle-propolis formulation exhibited the strongest radical scavenging capacity among all treatments, suggesting improved protection against oxidative stress in the wound environment. MTT assay results demonstrated a dose-dependent reduction in fibroblast viability at higher concentrations; however, no significant cytotoxicity was observed at low concentrations relevant to topical application (below ~0.78 ppm), and the nanoparticle-propolis formulation showed an acceptable biocompatibility profile compared to single-component formulations. Macroscopic wound healing results showed that healing rate and wound closure percentage in the nanoparticle-propolis group were significantly higher than in the control group, approaching near-complete healing by day 14. Histopathological analysis (H&E and Masson's trichrome staining) confirmed these findings, showing reduced inflammatory cell infiltration, improved epithelialization, increased

collagen deposition, and regeneration of skin appendages in the treated group. Overall, these findings suggest that the cyanobacterial TiO₂ nanoparticle–propolis formulation exhibits strong antifungal activity, antioxidant potential, and acceptable biocompatibility, and may be a promising candidate for further development as a topical agent for infected wound management. Future studies are required to investigate molecular mechanisms, long-term safety, and clinical applicability.

Data availability

Availability of data and materials: The datasets generated and/or analyzed during the current study are available in this published article.

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Author contributions

Conceptualization, B.N, F.B.; methodology, F.E.F.; software, B.N.; validation, B.N.; formal analysis, B.N. investigation, B.N.; resources, B.N.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics approval and consent to participate

Authors done experiments on mouse tissue samples. All experimental protocols and panelists involved in the study were accepted by ethics committee of Tehran medical sciences, Islamic Azad University, Tehran, Iran (IR.IAU.SRB.REC.1404.242). This study is an in vivo study on animals, and all experimental protocols were approved by the Research Ethics Committee of Islamic Azad University - Science and Research Branch (Approval ID: IR.IAU.SRB.REC.1404.242; Date of Approval: 2025-10-05).

Human and animal rights

Live vertebrates and/or higher invertebrates: This study is not on live vertebrates and/or higher invertebrates. All methods were conducted congruent with relevant guidelines and regulations as described by the ARRIVE guidelines. We confirm that all methods were performed in accordance with the relevant guidelines and regulations by including a statement in the methods section. The mice were sacrificed by Open-drop method of isoflurane. This method used to anesthetize mice according to the American Veterinary Medical Association (AVMA) Guidelines for the Euthanasia of Animals (2020). All the mice has been purchased from Iranian Research Organization for Science and Technology (IROST). We declared that we don't used human participants in the study.

Informed consent

All the images are original and not copied from other papers. A clear statement on informed consent obtained from all the panelists.

Additional information

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